

A1

Information Presentation Response

1. **What is something you learned from the presentation that you didn't know before?**
 - I learned that ACM Code of Ethics including the principles of enhancing public awareness and understanding of computing technologies and their social influence highlights an important and proactive role for professionals in public education.
2. **What area would you like to see explored in more detail in a follow-on presentation?**
 - I want to explore more on how IEEE and ACM code address emerging technologies, such as AI and blockchain, especially under the background of ethical dilemmas and practical case studies.

Ethical Advice

Advice for Sandra

- Since Sandra is the president of Med-E-Clinic Systems, she faces the biggest challenge among the three. She must balance the ethical concerns of her employees, the financial survival of the company and her responsibility of clients. Her central dilemma revolves is how to handle theses two controversial clients while maintaining company morale and adhering to professional standards. She must take care of this issues and ensure that her actions are both practical and ethically sound.
- First of all, Sandra must realize that she has the duty to uphold the company reputation for ethical behavior. ACM and IEEE Codes of Ethics provide the essential guidelines for this regards. ACM standard emphasized the avoiding the harm, acting in public interest and being honest and trustworthy. In the meanwhile, IEEE echoes these principles and stress the fairness, integrity and prioritizing public welfare. According to that, the clients Sandra took should suit for these standards to maintain the trust of her employees and public.
- The key issue Sandra faced is that if the message of providing software to an abortion clinic by Med-E-Clinic Systems is known widely, the company's reputation might face some challenges. Although abortion is legal in Canada, it remains a polarizing issue. Sandra must consider the potential backlash from stakeholders who might perceive this action as endorsing abortion. Likewise, homeopathic clinics also have raised ethical questions for promoting unscientific treatments, which may cause some damage with patients who used conventional medical treatments. These repetitional risks must be carefully considered with the financial imperative in this sales.

- According to this, Sandra need to develop some clear policies to address these challenges. These policies should outline the company stance on ethical problems and give the framework of future decision. It should stress the fairness and transparency, ensuring decisions are making consistently and based on sound reasoning rather than individual biases. This approach will help maintain the morale and public trust between employees.
- In order to solve this issue, Sandra must facilitate open and respectful communication between the team. The meeting described in article degenerated in personal attack and unproductive argument also display the importance of the need for better conflict resolution practices. Thus, Sandra should organize a additional meeting to solve the conflict between Sam and Cheryl. This meeting should be structured to encourage constructive dialogue and ensure that all voices are heard. Sandra must also set clear expectations for professional behavior, emphasizing that personal disagreements should not escalate into hostility.
- In the meanwhile, Sandra should pick on Sam's suggestion to sabotage the clients by installing buggy software. Such actions violate the professional ethics, harm the company's reputation, and potentially lead to legal consequences. Sandra should make this topic clear that any employee who engages in such behavior will face disciplinary action. Through the firm stance on such issues, Sandra can strengthen the company promise on ethical practice.
- In conclusion, Sandra must navigate such situation with focusing on ethical principles, transparent communication and practical decision making. Through consistently upholding professional standards with addressing employees concerns, she can lead her company through this challenge and set a positive example for ethical leadership.

Advice for Sam

- Sam's situation is tied with his personal beliefs and values. He disagree with the cooperation with abortion clinic from his moral convictions, which must be respected in some extent. However, he wanted to break such cooperation through installing buggy software is highly unethical, unprofessional and even illegal. He must realize that his beliefs can not give him the power to harm the company's reputation or its clients.
- ACM and IEEE code of ethical have provide the clear guidance on how professionals should do in such situations. The ACM Code stress avoiding harm, being honest, integrity and prioritizing the public good. Sam's proposal violated these principles. This action not only damage the clinic's operation but also tarnish the Med-E-Clinic Systems' reputation and expose the company to legal liability.
- Sam must realize that, as a professional, he should make his beliefs apart from the job responsibility. Although he has his own power to express his concern and apply for reassignment, he can't undermine his company or sabotage their clients. If Sam strongly

disagree with this cooperation, he should tell Sandra and request for reassigning to the other assignments. This approach allows him to maintain his moral integrity without compromising his professional responsibilities.

- Sam also need to understand that his action will cause what affects for his company or colleagues. The personal attacks he did to Cheryl during the meeting was also unprofessional and contributed to the breakdown of the communication within the team. He should apologize for his action to Cheryl and make effort to establish constructive work relationship. Professionalism required respecting for different opinion and cooperation, even though the group members may have different beliefs.
- It is also significant for Sam to consider the broader implications of his proposal to sabotage the abortion clinic. If such action have been discovered and spaced by someone, it may bring a inestimable damage for his company, not only lose the trust of clients but also need to face the legal action, and financial penalties. This consequence will harm not only the company but also its employees, including Sam himself. By prioritizing ethical behavior and professional integrity, Sam can avoid these risks and contribute to the company's success.
- Finally, Sam must learn to refuse all actions which may violate the ethical standard or company's clients. He should be able to convey his concern with respect attribute, if necessary, applying for change the position, and focus on keep professionalism in his interaction with clients or colleagues. By doing so, Sam can uphold both his personal values and his professional responsibilities

Advice for Cheryl

- Cheryl is opposition to working with the abortion clinic because she believes that homeopathy is unscientific and potentially harmful. Although it is good to concern about the ethical implication of promoting homeopathy, Cheryl must balance this concern and her professional responsibility. However, it is unethical that she agree with Sam's proposal because it undermines her credibility as a professional.
- ACM and IEEE code of ethical has provided the framework for solving Cheryl's dilemma. ACM standard emphasizes acting in the public space, honesty and avoiding damage. Similarly, the IEEE Code highlights the importance of fairness, transparency and the public welfare. The willingness of Cheryl to sabotage the homeopathic clinic directly violated these principles and will harm the clinic and its patients, damage the reputation of Med-E-Clinic System and expose the company into legal risks.
- Cheryl should realize that her personal thought for the homeopathic can't be a reason for doing unprofessional action. As a professional, her main responsibility is to offer high-quality service for her clients. If she feels strongly about not working with the homeopathic clinic, she should talk to Sandra and express herself to request reassignment. Only in this

way can she maintain her integrity without compromising the company's operations or violating ethical standards.

- In the meanwhile, Cheryl should consider the boarder context of her job. Although she might not agree with the practice of homeopathic clinic, her responsibility is to offer software product which helps them manage operation efficiently. Cheryl should recognize the cooperation between the clinic and her company is fulfilling her professional responsibility as a software developer instead of supporting the practice of homeopathic clinic. By focusing on the technical aspects of her work and ensuring that the software meets the client's needs, Cheryl can maintain her professionalism while addressing her concerns through constructive dialogue.
- Finally, Cheryl must refused the Sam's proposal which willing to sabotage the homeopathic clinic's software. This action will not only harm the clinic, but also damage to Cheryl's professional reputation and make her have the possibility to face the potential legal consequence. By adhering to ethical principles and prioritizing the public good, Cheryl can find a way, which suits for her values and professional responsibilities, to face this situation.
- In Summary, Cheryl must balance personal belief and work responsibility, use respectfully way to convey her concern, focus on offering high-quality service and reject unethical proposal. Keeping on these action, Cheryl can maintain her integrity and make contribute to the success of Med-E-Clinic Systems.

Report on a Software Disaster

Telangana exam result suicides

Introduction

- The Telangana exam result suicides of 2019 are a typical tragic example of how software error cause the destroyed real-world consequence. Over 26 students in Telangana, India, took their life after their academic performances from the Telangana State Board of Intermediate Education (TSBIE) exams were erroneously reported as failing grades. This incident was caused by several factors such as technical glitches, poor software practice and inadequate testing protocols and in the meanwhile, this incident simultaneously exacerbated the emotional and social stress faced by students and their families.

The Disaster

- The incident was caused by the software used to process and report the exam consequence. A private firm was responsible for handling this significant task but they haven't ensure the accuracy and reliability of the system. The wrong algorithm and improper data handling resulted numerous error in students' academic marks, such as

marking the passed students as failed. Besides, this software lacked completed validation mechanism to cross-check the processed data leading to report large number of incorrect data. TSBIE also didn't conduct adequate audits or quality checks before publishing the results. This unresopinsible action make the situation even worse than before. The consequence was dire -- students, who received their marks and unable to bear the stigma and pressure of failure, choose to take their live, while parents and the public demanded accountability from both the education board and the software vendor.

Key Issues

- As mentioned above, there are several reasons to bring about this event. To begin with, the education board haven't screened the private company carefully and allow them to use the substandard software to process the data. Secondly, they also haven't cross-checked the data generated or carry out the validation mechanism which leads to a tremendous amount of errors. Thirdly, the lack of clear contingency plans or error-handling routines delayed the correction of mistakes. In addition, students' emotional and pressure-bearing capacity are weak, which showed that schools do not have mental health support systems.

Lessons Learned

- This incident emphasizes the need for tighter controls in outsourcing highly sensitive tasks, such as the processing of examination results. There is a need for government agencies to ensure that the contractors are strictly following the best standards in developing and testing software. Sensitive data such as the results of students must go through obligatory detailed audits and validation processes before release. Furthermore, educational institutions should provide mental health services and reinforce the idea that their self-esteem does not depend on academic performance. Additionally, quick and efficient crisis management procedures must be planned and prepared for when incidents of this type occur to reduce potential harm.

Conclusion

- The Telangana exam result suicides provide a stark reminder of the far-reaching consequences of software errors, particularly where they touch on high-stress situations like student examinations. The issue needs to be tackled from many fronts: improving software quality assurance, strengthening mental health support systems, and fostering a culture where students are made to feel worth more than their grades. By learning from these incidents, governments and organisations can prevent such tragedies from occurring in the future and ensure that technology serves for empowerment, not otherwise.

TUT Left Question:

- In the classic trolley problem, I would never insert myself and pull the lever under any circumstance, unless someone I know is on one of the tracks or there is a group of people explicitly issuing a clear command for me to do so. Under these extraordinary circumstances, I might well intervene and pull the switch. Here are my reasons:
 - When I decide to do something, I become personally accountable for the result, either good or evil. Where there are marginal cases where I am not lord over the outcome, it is better to avoid doing something that will have uncertain effects.
 - Barring specific consent or directive from the parties directly concerned, my opinion is that it is morally wrong for me to have my option imposed on other people's lives where the stakes happen to be as high as life and death.
 - The psychological cost of actually taking the lives of people, even though it will result in saving more lives, can be devastating. Being a bystander lowers the moral discomfort of choosing who remains alive and who dies.
 - Even when the circumstances take on a definite binary result, in most real-world scenarios, there is doubt. Acting on partial knowledge can inadvertently create more harm than taking no action.
 - If a loved one is on one track, my personal relationship and desire to protect them would override my general rule of non-interference. And if a group of people quite obviously calls for action, their command gives moral clarity and diverts some of the blame from me.